

WASP-135b

R-1165

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_220426 · created 2026-07-31T02:47:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459695.7748 +/- 0.0074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.05
Transit depth (Rp/Rs)^2	2.96 +/- 1.73 [%]
Orbital Inclination (inc)	80.56 +/- 2.79
Airmass coefficient 1 (a1)	1.02 +/- 0.031
Airmass coefficient 2 (a2)	-0.006 +/- 0.01
Scatter in the residuals of the lightcurve fit is	3.07 %
Best Comparison Star	#3 - [453, 318]
Optimal Method	PSF photometry
Transit Duration (day)	0.042 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.172 ± 0.05 vs 0.139 (deviation 0.66σ)
Duration fit vs published	0.042 d vs 0.0687 d (deviation 1.22σ)
Mid-time O–C	-11.3 min
Depth SNR	3.4
Residual scatter	3.07 %

Light curve

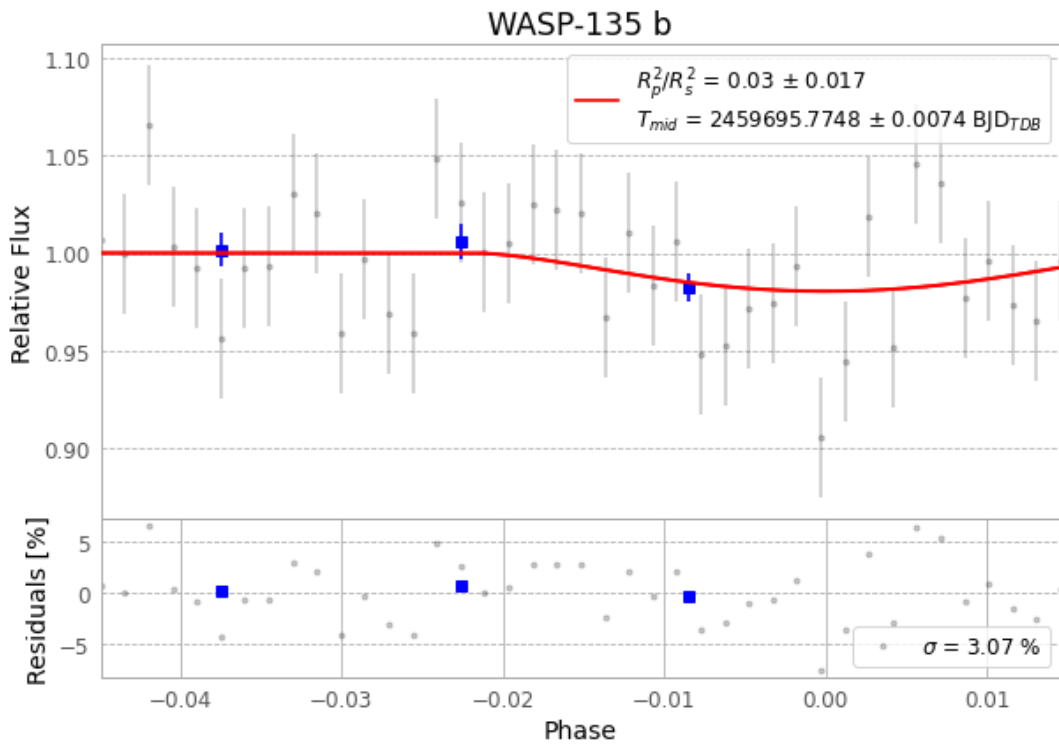


Figure 1 — FinalLightCurve_WASP-135 b_2022-04-25.png (SHA-256 345021d43efeb5b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [171, 316], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d66ed65bb36979a9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

42 FITS frames (28 MB), WASP-135220426045714.FITS → WASP-135220426070013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-220426.log (a8caca7650b1...), FinalLightCurve_WASP-135 b_2022-04-25.png (345021d43efe...), FinalLightCurve_WASP-135 b_2022-04-25.csv (0158c5c343d5...)

Compute resources

Wall time 80.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 02:47Z.